

AMARDEEP SAHA

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LeetCode | GitHub | LinkedIn | Portfolio

Built a distributed Intent-to-Lead pipeline processing **2.15M+ buyer clickstream events/day**, using **Cassandra** for high-throughput activity logging, scheduled ETL for extracting buyer search interests and enriching product/seller metadata, and **Kafka**-based asynchronous processing for downstream lead creation; converted **~14%** of qualified intents into sold leads.

Designed and implemented a WhatsApp-based enquiry generation pipeline that identified user intent for specific modules and triggered automated engagement, resulting in **INR 1 lakh additional profit**, scaled onboarding by enabling **200+ additional users per day**.

Architected and implemented a high-performance concurrent caching layer in **Golang** leveraging in-memory storage and **Redis**, reducing database calls by **90%+** and achieving **~1ms p95 latency** with a **3-second warm-up** and near-zero-latency steady-state reads.

Led end-to-end full-stack performance optimization initiatives by reviewing **5+ service SRS documents**, identifying lightweight backend API strategies, and transforming sequential service interactions into asynchronous workflows, resulting in a **25% reduction** in overall page load time.

Designed and developed a **Kafka**-driven location validation and enrichment pipeline in **Golang** with **PostgreSQL**, detecting location mismatches exceeding **250 km** and triggering asynchronous WhatsApp-based remediation workflows to improve data accuracy and automation.

Implemented Server-Side Rendering (**SSR**) using **React** and **Node.js** to enhance SEO, improve search engine indexability, and drive a **5% increase** in organic traffic.

Engineered scalable, data-intensive frontend architectures capable of handling **200+ concurrent asynchronous requests** per component while maintaining UI responsiveness, stability, and performance under high load.

Owned and optimized automated data enrichment workflows for mobile user enquiries, increasing valid form fill rates by **10%** through intelligent extraction and enrichment strategies.

Architected a unified **JavaScript** versioning and deployment framework that eliminated cross-team release dependencies and streamlined the rollout of centralized shared components.

Designed and delivered a reusable UI component library comprising **20+ standardized components**, improving design consistency and accelerating feature development across multiple projects.

Collaborated proactively with cross-functional product, backend, and platform teams to drive rapid iteration cycles, continuous improvements, and efficient delivery of high-impact features.

Spearheaded the implementation of Google's **one-click unsubscribe** compliance feature for promotional emails, coordinating execution across multiple teams to ensure seamless adoption and regulatory alignment.

Integrated a sentiment analysis capability using HuggingFace's **multilingual BERT** model to evaluate user tone and intent, enabling smarter downstream processing and enhanced user interaction analysis.

Designed and orchestrated custom LLM nodes for semantic lead-matching using **LangGraph** and **LangChain**, with **Langfuse** for observability; combined prompt compression, request batching, and flex processing to cut per-request inference cost from **INR 0.80 to under INR 0.10**, sustaining **10K+ requests/day** with response latency **under 10s**.

Built a concurrency-safe lead-routing pipeline that curates data from **10+ APIs** and routes qualified sellers to users over WhatsApp, using **Redis** distributed locks and atomic **INCR** transactions for race-free processing across distributed goroutines; polled curation status with a **5s** network cap against a **60s SLA**, returning results as soon as all source APIs resolved (**~92% reduction** in worst-case network hold).

EDUCATION

Indian Institute of Information Technology , Ranchi
Computer Science and Engineering B.Tech.
CGPA: 8.45

Ranchi, India
July 2019 - May 2023

EXPERIENCE

Indiamart | Full-Stack Developer

Noida, India | June 2023 - Present

CoantumLeap | Research and Development Intern

Remote, India | March 2022 - August 2022

- Engineered a full-stack DAO application to manage over 10,000 members and transactions.
- Optimized gas-efficient smart contracts in Solidity for token sharing and stake distribution logic.

- Implemented frontend-to-blockchain connectivity using Ethers.js and MetaMask.
- Deployed smart contracts to the Ethereum blockchain with Remix IDE.

SKILLS

Programming Languages:	Golang, Python, JavaScript, TypeScript, SQL, Solidity, C++, C, PHP, Bash, HTML
Backend & Frameworks:	Gin, Node.js, Express, REST API, Apache Kafka, Redux, Next.js, React, Yii2
AI / LLM Tooling:	LangChain, LangGraph, Langfuse, HuggingFace Transformers, Groq SDK, Prompt Engineering
Databases & Messaging:	PostgreSQL, Cassandra, Redis, Apache Kafka, MongoDB, SQLite
Tools / Platforms:	Git, Docker, Devtron, Windmill, VS Code, Remix, Ethereum

PROJECTS / OPEN-SOURCE

Video Conferencing App | Link *React JS, WebRTC , SocketIO, SQL , NodeJS, Redis, Docker*

- Engineered a full-stack, real-time video conferencing application .
- Implemented peer-to-peer audio/video streaming with WebRTC and established real-time signaling and chat functionality using Socket.IO.
- Developed a secure backend featuring JWT-based authentication and a robust room management system, paired with a modern, responsive UI built with Tailwind CSS.

Voice-Enabled Task Tracker | Link *Next.js, TypeScript, Node.js, Express, SQLite, LLMs, Docker*

- Built a backend-driven task management system with REST APIs supporting CRUD, filtering, and search, using Node.js, Express, and SQLite with clear service separation.
- Designed an LLM-powered natural language processing pipeline using Groq SDK, converting voice input into structured task data (title, priority, due date, status).
- Integrated speech-to-text ingestion with backend services and a Next.js frontend, enabling real-time voice-based task creation and Kanban workflows.

Research Paper " A Survey Paper on Developments in Blockchain"

- Authored and published an accepted research paper, "A Survey Paper on Developments in Blockchain," at IIIT Ranchi.
- Conducted a comprehensive literature review, analyzing over 25 research papers to evaluate the evolution of blockchain consensus mechanisms.